

Instructor Guide

Low-Rank Structure Is Geometry - Paul Breiding and Se Eun Choi

Scope boundary. This curriculum does not present tensor decomposition as an alignment algorithm. It studies the mathematical conditions under which low-rank representations may become useful evidence in interpretability and alignment auditing.

The teaching arc



Do not let the class jump directly from a small reconstruction error to a semantic claim. The sequence deliberately makes learners identify the represented object, its coordinate symmetries, the model's structural assumptions, and the numerical evidence before discussing neural concepts.

Alignment evidence ladder

1 Reconstruction	The selected model approximates the selected data object.
2 Invariance	Coordinate symmetries and identifiability limits are accounted for.
3 Conditioning	Small perturbations do not create uncontrolled changes.
4 Stability	The finding persists across seeds, perturbations, ranks, and solvers.
5 Behavior	External examples, labels, or held-out tasks support the claim.
6 Intervention	Changing the feature produces the predicted effect under controls.

Key sentence: Good reconstruction alone stops at Step 1.

Before the session

1. Instantiate the pinned Julia environment and confirm `using TensorKitchen` succeeds.
2. Open notebooks once in Pluto to absorb compilation time before class.
3. Keep code folded and reveal each experiment only after a prediction.
4. Confirm the slide HTML and both exercise PDFs open locally.

Teach one claim at a time

Predict - Observe - Explain - Limit the claim

Resource	Core reveal	Question to leave open
Primer 10-15 min	Tensor modes; CP rank-one summands; Tucker subspaces and core.	What has low reconstruction error not established?
Lab 1 20 min	Matrix gauge, then CP rescaling and permutation.	Which displayed coordinate quantities are invariant enough to interpret?
Lab 2 15-20 min	One target viewed through CP, Tucker, and BTD model cards.	Does the component model match the proposed semantic claim?
Lab 3 25 min	Failure map, collision distance, ALS conditioning, three-case comparison, solver race, plateau microscope.	Does the plateau diagnose its cause, or is more evidence needed?
Lab 4 20-25 min	Patch activations, NMF concept recipes, rank vocabulary, example-based labeling, importance, recursion.	What stability and behavioral evidence would make the candidate concept defensible?

Reusable facilitation prompts

Predict. What should change and what should stay invariant?

Explain. Is the result about the object, coordinates, model, or optimizer?

Observe. Which visual or diagnostic tests that prediction?

Limit. Which evidence-ladder step has actually been reached?

Lab-specific emphasis

Lab 1 - invariance before interpretation

Have learners name the unchanged matrix before discussing the changing factors. Transfer the same reasoning to reciprocal CP scaling and component permutation.

Lab 2 - assumption before labeling

Require a sentence beginning "This component is interpretable only if ...". Separate structural capacity from whether a finite solver run reaches that structure.

Lab 3 - stability before trust

Connect the small Gram eigenvalue to difficulty assigning contribution between nearly identical components. Stress that the plateau flag records slow progress only; distance and conditioning are separate evidence used to diagnose its cause.

Lab 4 - a factor is a hypothesis, not a label

Have learners connect one candidate direction to several high-usage crops, then perturb its coefficient. Nonnegativity supports an additive reading but proves neither uniqueness nor semantics. Clearly label the sample $\times$ location $\times$ feature model as a TensorKitchen extension inspired by CRAFT.

Ask what the evidence warrants

Use the exercise sheet for observations; use the ladder for claims

Final group report

Ask each group to choose one notebook result and report five items:

1. the represented object;
2. the relevant coordinate equivalence;
3. the model assumption;
4. the conditioning or stability evidence;
5. the behavioral or intervention test still required.

Do not award full credit for “the reconstruction error is low” unless the question asks only about Step 1.

Common misconceptions

Misconception	Instructor response
The displayed factor is unique.	Ask which rescalings, permutations, or basis changes preserve reconstruction.
Nonnegativity proves semantic meaning.	Treat it as a structural assumption and request external validation.
The best finite-run error identifies the best model.	Separate model capacity, optimization success, and scientific meaning.
A plateau means convergence.	Compare recent improvement with gradient, conditioning, and component activity.
A stable feature is causal.	Stability is Step 4; intervention is Step 6.

Materials map

CURRICULUM.md	Rationale, prerequisite map, learning outcomes, and evidence ladder.
notebooks/00-04	Guided experiments in the intended learning order.
exercises/	Student Markdown, printable HTML, and six-page worksheet PDF.
instructor/	Editable teaching notes, short answer key, and this guide.
slides/	Interactive Pluto deck and standalone HTML export.
REFERENCES.md	Mathematical, AI, software, and teaching-source provenance.

Closing message. A low-rank representation can be useful audit evidence only when its mathematical ambiguity, numerical reliability, and behavioral meaning are stated together.